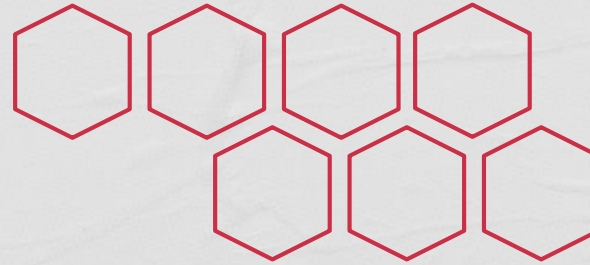


Safety Requirements On Storage Of Hazardous Chemicals



Prepared by: Waria Alghafri
Designed by: Mohammed Al Hadrami
Health, Safety and Environment, Royal Hospital



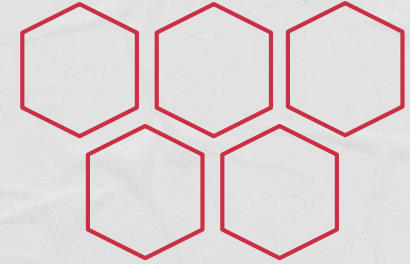
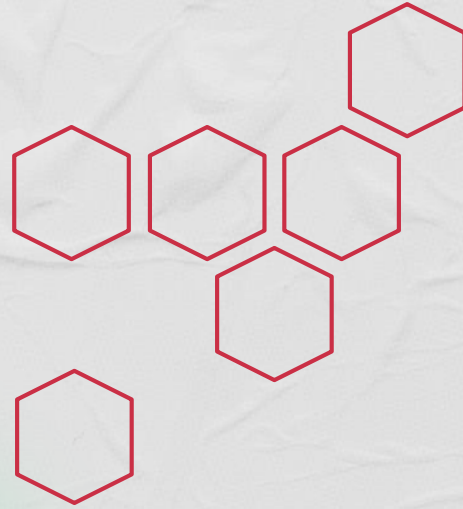


Introduction

A hazardous substance can be any substance, whether solid, liquid or gas, that may cause harm to your health.

Hazardous substances are classified on the basis of their potential health effects, whether acute (immediate) or chronic (long-term)





01

Physical States Of Hazardous Materials

Solid



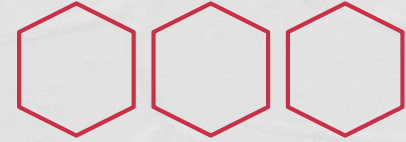
Gas



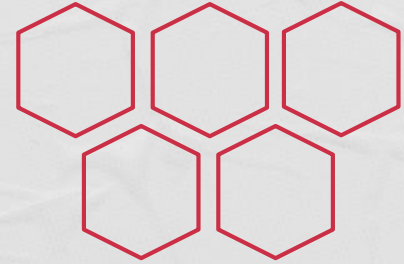
Liquid



vapor



HCS Pictograms and Hazards		
Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Reactives • Organic Peroxides	Exclamation Mark  • Irritant (skin and eyes) • Skin Sensitizer • Acute Toxicity (harmful) • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-mandatory)
Gas Cylinder  • Gases Under Pressure	Corrosion  • Skin Corrosion/burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides
Flame Over Circle  • Oxidizers	Environment (Non-Mandatory)  • Aquatic Toxicity	Skull and Crossbones  • Acute Toxicity (fatal or toxic)



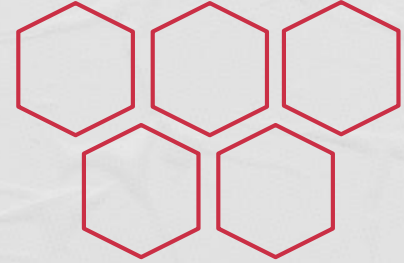
02

Categories Of Hazardous Chemicals



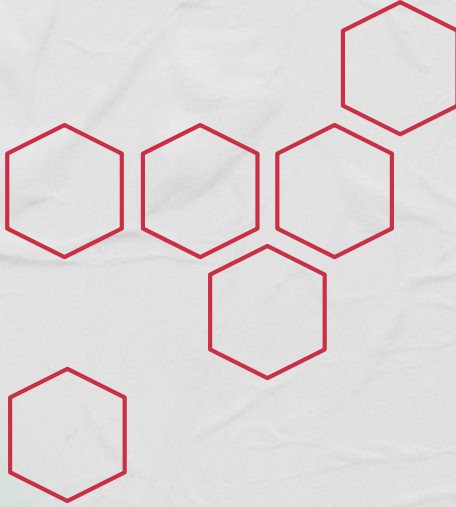
- | | |
|--------------|---------------------------------|
| 01 toxic | 06 carcinogenic |
| 02 harmful | 07 Causes specific organ damage |
| 03 irritant | |
| 04 mutagenic | 08 Respiratory sensitizing |
| 05 corrosive | 09 Toxic for reproduction |





03

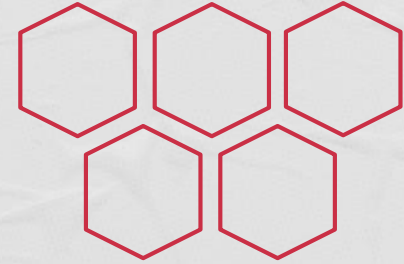
Who Is At Risk From Hazardous Substance



Anyone who deal or exposed to hazardous substances is at risk.

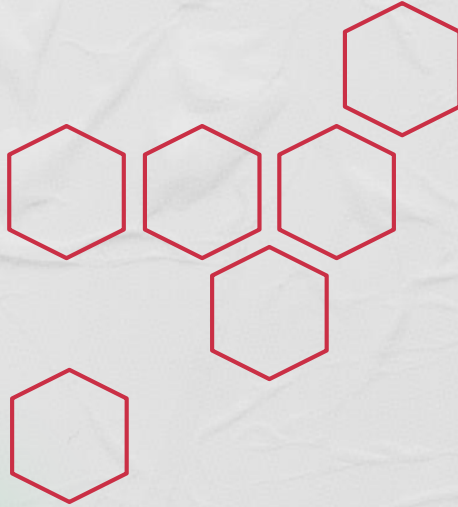
Those exposed to more hazards substances for long period of times are more at risk than those exposed for short periods to less hazards substances.

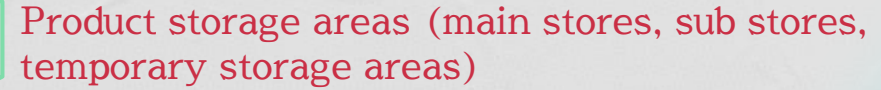
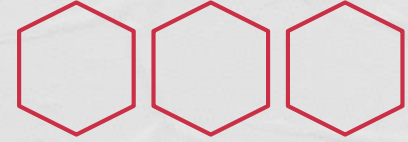
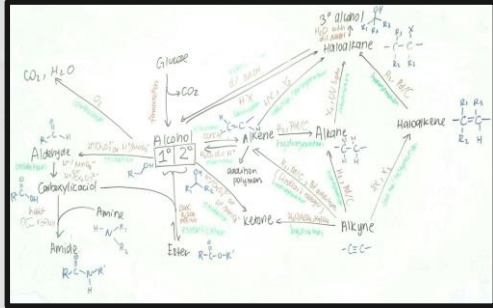


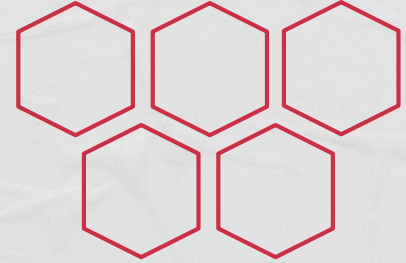


04

Flow Chart Of Chemical Storage

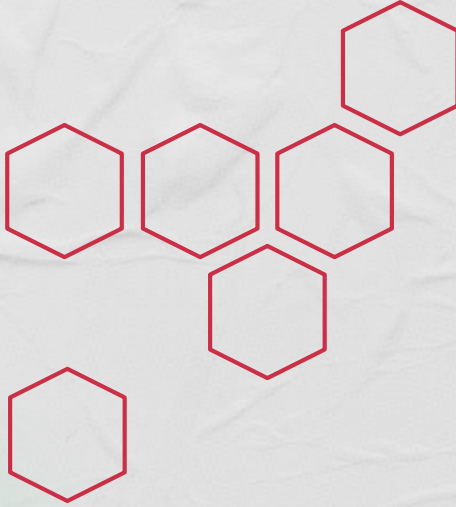


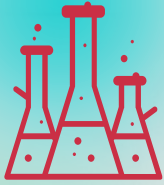




04

Precautionary Measures

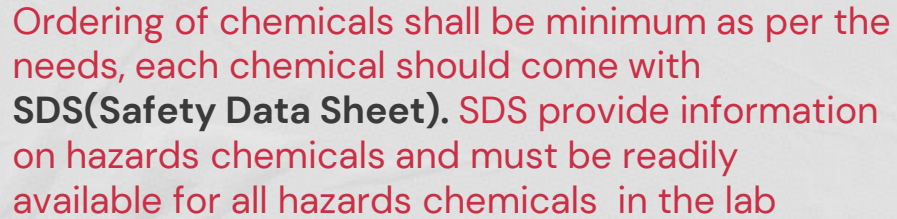




Shall be adopted to protect workers against the risks of exposure to the chemicals which leak to the work environment such as **gases** , **dust** ,**liquids** and **acids**

The following shall be taken into consideration

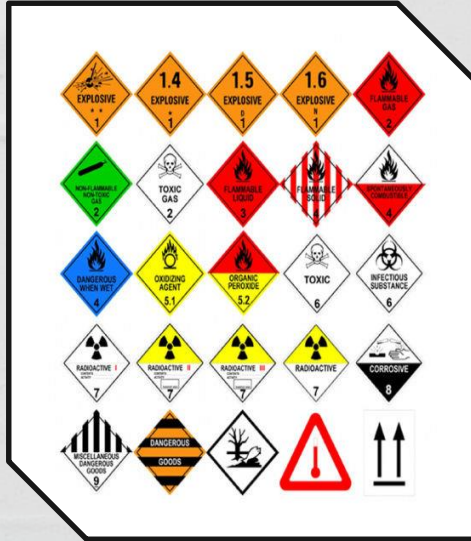






02

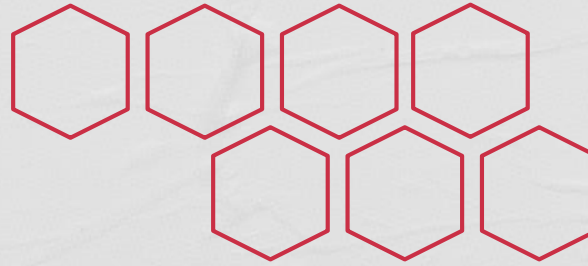
Storage room should be constructed with fire resistant material with a minimum fire resistance of two hours. It should be explosive proof suitable to keep highly flammable materials, well ventilated.



03

The storage compartments shall be **labeled** with in according to their hazards class.

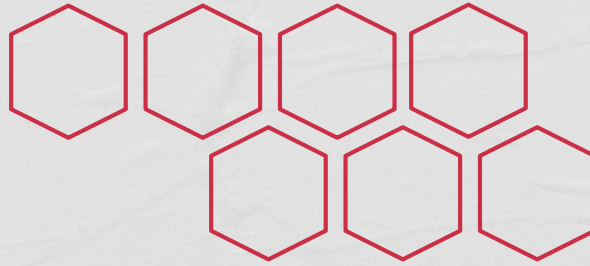




04

Do not store chemicals **near heat sources** such as ovens or steam pipes. Also, do not store chemicals in direct sunlight.

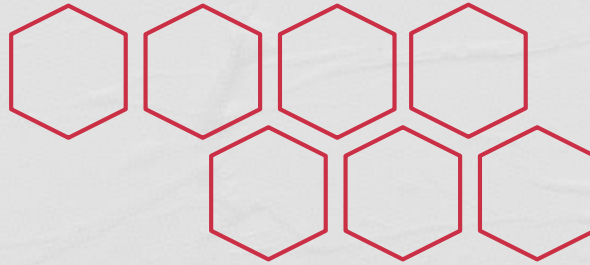




05

Liquid and solid materials are to be stored separately where possible ,to avoid contamination in case of spillage





06

Do not store any chemicals in glass containers above the waist level.





07

Inspect your chemicals regularly for any signs of deterioration and for the integrity of the label.



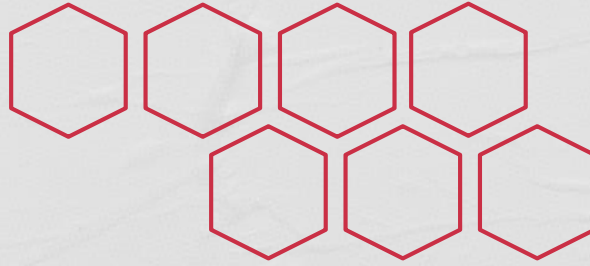
08

Eye wash and emergency shower facility
should be available



09

Equipped with **fire extinguishers** with suitable quantity and type (near escape route)



10

Smoke detection should be installed





11

Spill control **and** clean up materials **must be**
available **at all times**



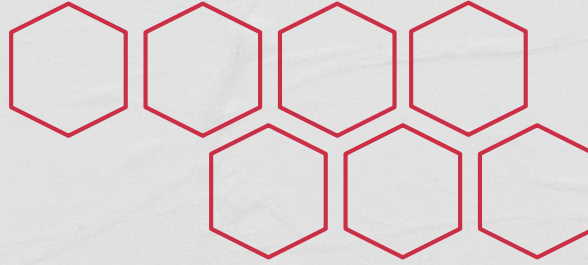
12

Smoking, eating or drinking in the store
or during handling of chemical substances
are prohibited



13

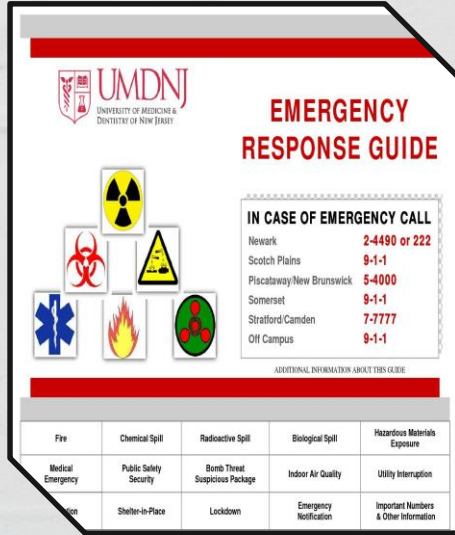
Appropriate decontamination procedures , such as **washing hands** with soap and water after handling chemicals in the store **must be followed**.



14

Empty containers are to be disposed of with other waste .to prevent re-use for unauthorized applications they need to be removed **with other chemicals waste** ,according to the chemical waste disposal guidelines.





15

Emergency instructions **must be prominently posted**

Caution
MCN – TT 0110
OCRS Chemical Safety

Emergency Contact Information
John Doe
Day – 615-322-2057, Emergency – 615-875-3779

Emergency Information



Regulatory Information



PPE/Precautions



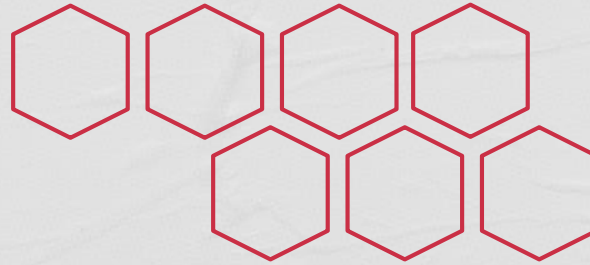
Area Hazards



Date created: 07/02/2021

16

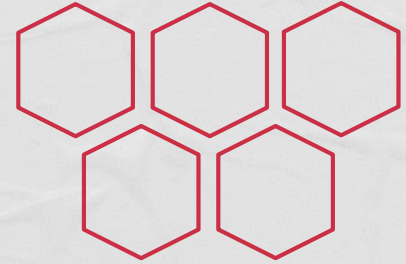
Laboratory emergency **contacts** must be posted outside each work area



17

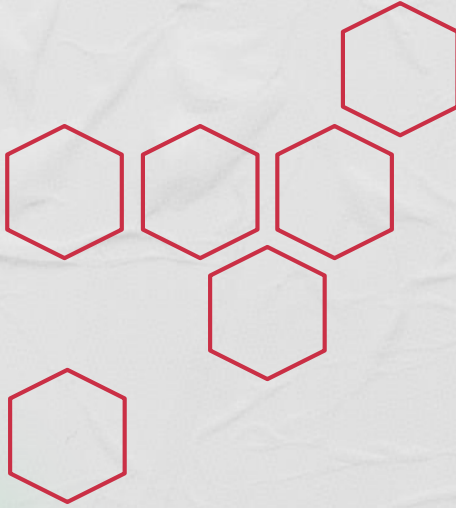
Appropriate **personal protective equipment**
should be available at all time





05

How To Handle Chemicals Properly



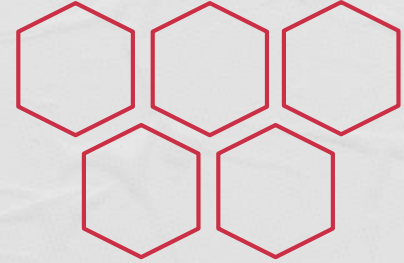
Use cautions

Always follow procedures

Read all labels

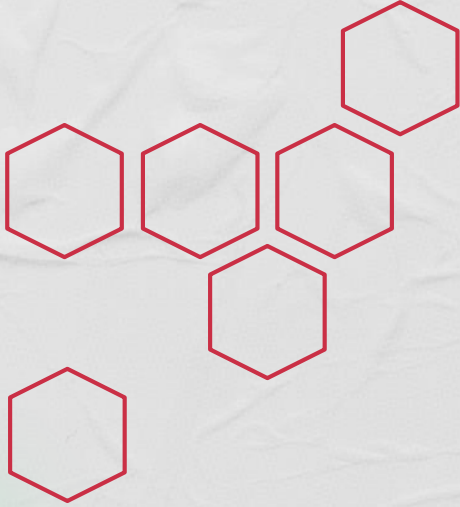
Keep yourself and work area clean





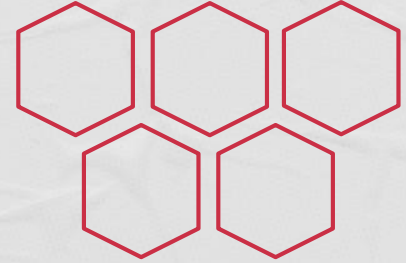
06

Other Safety Requirements



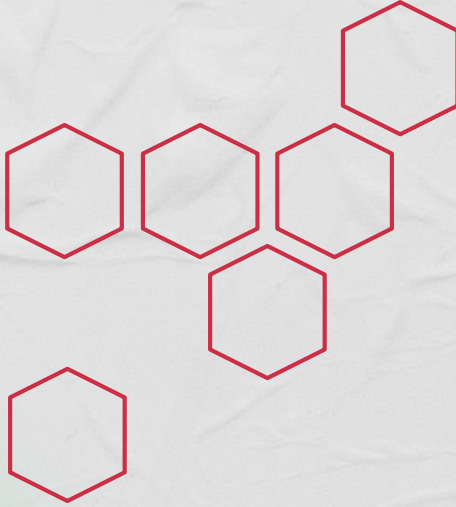
- ❖ **Eye wash and emergency shower facility should be available**
- ❖ **Spill control and clean up materials must be available at all time .**
- ❖ **installed CCTV system in the store and connecting with control room**
- ❖ **Volatile or or flammable toxic substances should be stored in ventilated storage cabinets.**





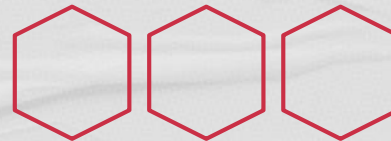
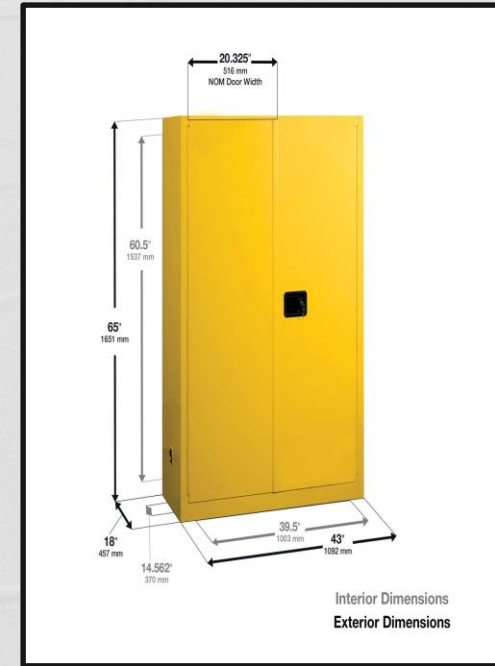
06

Safety Cabinet



Safety specification of chemical storage cabinet:

- ⬡ **Eye Designed for the safe storage of chemicals and flammable liquids**
- ⬡ **Storage capacity(as per requirement)**
- ⬡ **Fire-resistant, leak-tested design**
- ⬡ **Manual close(better self closing)**
- ⬡ **Warning Label**
- ⬡ **Complies with OSHA, NFPA standard**

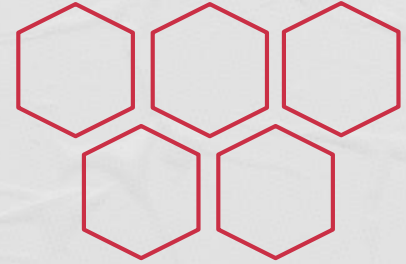


Request safety cabinet as per your needs
and determine the following:

- ⬡ **Cabinet body** Double fireproof steel construction
- ⬡ **Color** Easily identify, organize and segregate liquids by using the correct color cabinets.
- ⬡ **Size and type of storage cans** Determine the length and width
- ⬡ **Capacity** Select specialty cabinets for on-the-spot needs larger cabinets for expanded or large quantity storage.
- ⬡ **Numbers of door**
- ⬡ **Total Number of Shelves**

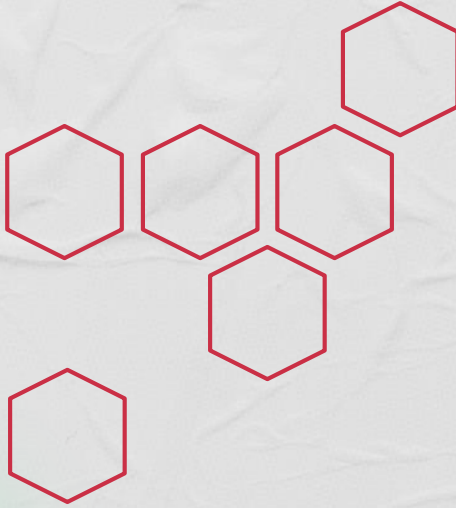
POINT-OF-USE STORAGE REQUEST FORM		Page 2 of 2	
If any of the following questions are answered "No," additional sheets outlining the proposed procedure must be attached.			
General Criteria:		Y	N
Has a map showing the locations of all flammable and corrosive cabinets (including this POU area) that are bounded within the fire zone been attached to this application?			
If there are multiple cabinets within the fire zone, have the volume and type of chemical in each cabinet been indicated on the map?			
Environmental Criteria:		Y	N
Will the volume of each chemical used be recorded in the hazardous materials management system at least 3 times per week?			
Will each chemical stored at the POU area be directly attributed to a single process code?			
Is the current issue method a hazardous material pharmacy?			
Are there procedures to ensure that no more than a 2-day supply of chemicals is maintained at the POU area?			
Are there procedures to check shelf life expiration for each chemical at the POU area?			
Is there a stocked, serviceable spill kit within 25 feet of the proposed POU area?			
Will the existing IAPs be used with no charges to the waste descriptions or additional drums needed?			
Safety Criteria:		Y	N
Do the flammable storage cabinets conform to OSHA and NFPA 30 Standards?			
Do the corrosive storage cabinets conform to appropriate safety standards?			
Location/area was inspected to ensure all safety requirements are met (e.g. electrical hazards, operational compatibility, etc).			
Fire Protection Criteria:		Y	N
Including the chemicals at this POU area, will the total volume of Class I, II and IIIA chemicals be less than 360 gallons in the fire area?			
Will the total volume of Class I, II, and IIIA liquids stored in any one cabinet be less than 120 gallons?			
Will the total volume of Class I and II liquids stored in any one cabinet be less than 60 gallons?			
Are the cabinets in the fire zone, including the cabinets for this POU, numbered and are the numbers indicated in the attached map?			
Is there an ABC fire extinguisher within 50feet of the proposed POU area?			
Is there a procedure in place to ensure that flammables are returned to the cabinets at the end of each shift?			
Concur/Nonconcur—See attachment _____		Concur/Nonconcur—See attachment _____	
Environmental Management _____		Safety _____	



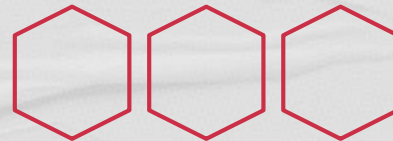


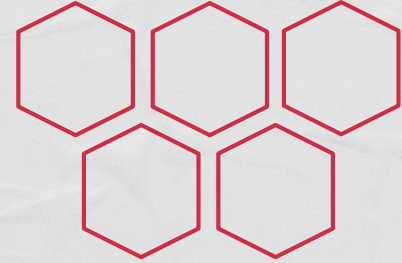
07

Spill Pallet



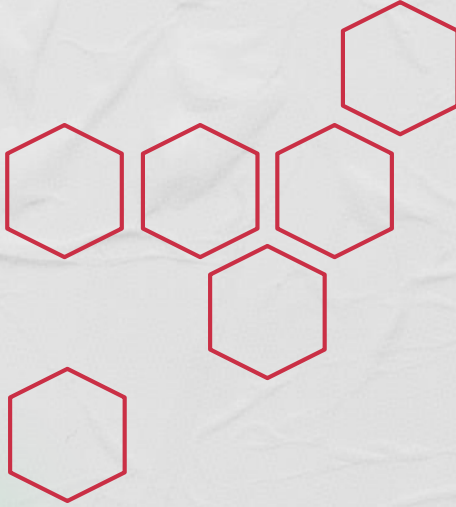
- ⬡ highly chemical resistant, compatibility chart attached
- ⬡ easily removable heavy duty grates.
- ⬡ low profile design for easy loading.
- ⬡ color coded yellow for easy visibility.
- ⬡ tested and proven load capacity ensures additional safety.
- ⬡ anti-rust, uv resistant, anti-corrosion, & anti skid grates.





08

CHEMICALS TRANSPORT TROLLEY



- With stainless steel framework
- Polyethylene Spill tray with zinc plates steel handle
- Transports 4 no. 25 liter Drums



THANKS

Do you have any questions?

w_asal@hotmail.com

9926 / 9216

HEALTH ,SAFETY AND ENVIRONMENT DEPARTMENT

